

Electrodynamics

Independence Semester 2024

Homework 4

November 4, 2024

Problem 1

a) Suppose $G_N(\vec{r}, \vec{r}')$ is a Neumann Green function in a volume V bounded by a surface S , and that it is chosen to obey $\partial G_N(\vec{r}, \vec{r}')/\partial n' = -4\pi/A$ when $\vec{r}'$ lies on S . Here, $A = \int_S dS$ is the total area of the surface S . Show that by applying Green's theorem with $\phi(\vec{r}) = G_N(\vec{r}_1, \vec{r}')$ and $\psi(\vec{r}) = G_N(\vec{r}_2, \vec{r}')$ does not allow one to show that $G_N(\vec{r}, \vec{r}')$ must be symmetric in $\vec{r}$ and $\vec{r}'$.

b) Show that it is possible to define a new Neumann Green function $\tilde{G}_N(\vec{r}, \vec{r}') = G_N(\vec{r}, \vec{r}') + F(\vec{r}, \vec{r}')$, which obeys the same Poisson equation $\nabla'^2 \tilde{G}_N(\vec{r}, \vec{r}') = -4\pi\delta^3(\vec{r} - \vec{r}')$ as $G_N(\vec{r}, \vec{r}')$ does, and which satisfies the same boundary condition, and that is symmetric in $\vec{r}$ and $\vec{r}'$. Find $F(\vec{r}, \vec{r}')$ explicitly.

Problem 2

Please recall that the Robin boundary condition is given by, $\left(\frac{\partial\phi}{\partial n} + a\phi\right)\Big|_S = f_R$, where f_R is a freely specifiable function on the boundary, and a is a strictly non-negative but otherwise freely specifiable function on the boundary. Using the same procedure as we used in the class to prove the Uniqueness theorem for Dirichlet and Neumann boundary conditions, prove the uniqueness theorem for the Robin boundary condition.

Problem 3

Please consider a conducting sphere a is held at zero potential, and a charge q is placed outside the sphere, at a distance b from the origin. The situation can be 'mocked up' by considering instead entirely free space containing the original charge and also a certain

charge q' placed at a certain distance c from the origin of the sphere, on the line joining the charge q and the origin.

a) First, for simplicity consider the sphere is centred on the origin of Cartesian coordinates, and that the charge q is placed at distance b along the z -axis, i.e., $(x, y, z) = (0, 0, b)$. Now, assume that the image charge q' also lie on the z -axis, i.e., $(x, y, z) = (0, 0, c)$. We need to show that this image charge together with the original charge indeed give rise to a spherical surface of radius a that has zero potential. Assuming that the above arrangement gives rise to zero potential at the north pole $(0, 0, a)$ and south pole $(0, 0, -a)$ on the sphere, find the expressions for q' and c in terms of a , q , and b .

b) Next, using the results of part a) show that the potential is zero at a point P at an angle θ from the z -axis. Why is it sufficient to prove that the potential is zero at the above mentioned point P , to prove that the potential is zero on all points on the spherical surface of radius a .

c) Now, you need to rewrite the results of part a) in a more general fashion, in which the original charge q is placed at an arbitrary point $\vec{r}_1$ located outside the sphere, rather than lying specifically on the z -axis.

d) Using the results of part c), write down the potential $\phi(\vec{r})$ at $\vec{r}$ outside the zero potential sphere at $r = a$ due to a charge q located at $\vec{r}_1$ outside the sphere.

e) If γ is the angle between $\vec{r}$ and $\vec{r}_1$, find the surface charge density on the conductor by using $\sigma = \frac{\hat{n} \cdot \vec{E}}{4\pi}$.

f) Also, using the results of part d) , construct the Dirichlet Green function for the boundary-value problem where the potential is specified on the surface of a sphere.

g) Assuming $\rho = 0$ find out $\phi(\vec{r})$ from

$$\phi(\vec{r}) = \int_V \rho(\vec{r}') G_D(\vec{r}, \vec{r}') d^3\vec{r}' - \frac{1}{4\pi} \int_S \phi(\vec{r}') \frac{\partial G_D(\vec{r}, \vec{r}')}{\partial n} dS'. \quad (1)$$

Problem 4

Construct the Dirichlet Green function $G_D(x, y, z; x', y', z')$ for the infinite-volume region comprising the positive octant in Euclidean space, i.e. the region where $x > 0$, $y > 0$, $z > 0$. (This Green function can be used in order to solve the Dirichlet boundary-value problem in the positive octant, where the value of the potential is specified on the three orthogonal planes $x = 0$, $y = 0$ and $z = 0$.)

[*Hint:* first use the method of images to find the potential in the positive octant due to a charge q at $\vec{r}_1 = (x_1, y_1, z_1)$, in the presence of three orthogonal grounded planes at $x = 0$, $y = 0$ and $z = 0$.]

b) By making appropriate changes to the signs of the coefficients of the various terms you obtained in the expression for the Dirichlet Green function in part a), construct the Neumann Green function $G_N(x, y, z; x', y', z')$ for the positive octant. Please demonstrate explicitly that your result for $G_N(x, y, z; x', y', z')$ does indeed satisfy the required boundary conditions.

Problem 5

Consider a two-dimensional potential theory problem in the (x, y) plane, where the solution to $(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2})\phi(x, y) = 0$ is required inside the semi-infinite U-shaped channel with sides $y \geq 0$ at $x = 0$; $y \geq 0$ at $x = a$; and $0 \leq x \leq a$ at $y = 0$. Suppose $\phi(x, y)$ obey the Dirichlet boundary condition that $\phi(0, y) = 0$ and $\phi(a, y) = 0$ for all $y \geq 0$; and that $\phi(x, 0) = V(x)$ for $0 < x < a$, where $V(x)$ is a function to be specified later. Assume that $\phi(x, y)$ goes to zero as y goes to infinity.

a) Use the method of separation of variables to construct the solution for $\phi(x, y)$ inside the U-shaped channel, as an infinite series with coefficients expressed in terms of integrals involving $V(x)$.

b) Obtain the solution explicitly, in closed form, if $V(x) = V_0 \sin \frac{3\pi x}{a}$, where V_0 is a constant.